CS471 – Web Technologies (Laboratory)	Qassim University College of Computer	Lab 8
		Django Models (Part 2)

This lab session covers Django Models (Part 2 – Q operator, sorting objects, aggregation functions, and multi-models). By the end of this session, students should have a very good understanding of how to build advanced queries with multi-model query in Django.

Pre-lab Preparation:

1. The table book as in lab7.
2. The fake data from mockaroo.com website or any other methods.

Lab Activities:

Task 1: Create a URL '/books/lab8/task1' with any necessary HTML file and view function to list books that have price less than or equal 80 using Q operator.

Books with price less than or equal to 80

Title	Author	Price	Edition
Continuous Delivery	J.Humble and D. Farley	0.0	1
Good Son, The	Elnore Woodroff	19.99	69
Vizontele	Hastings Baumber	24.99	780
Love, Wedding, Marriage	Heath Lauxmann	25.0	166
Hothead (Coup de tête)	Keslie Ivashkin	2.49	389
Lawyer Man	Rancell Wissbey	39.99	694
The Big Bus	Malvina Lumbers	24.99	13788
Things to Do in Denver When You're Dead	Vernon Brixey	6.29	64
Oh, God! Book II	Elayne Mantle	3.79	99
Applause (Applaus)	Von Lagen	1.99	1
Curious Case of Benjamin Button, The	Estevan Kleinzweig	19.99	739
Faculty, The	Dorey Fraulo	5.49	40457
Beauty in Trouble (Kráska v nesnázích)	Rockey Doucette	8.99	74
Carousel	Mordy Skingle	24.99	285
Perfect Blue	Marieann Liston	69.99	2
Midnight Movie	Miquela Struys	22.99	7
Sahara	Hersch Pykett	4.49	4
We Are Young. We Are Strong.	Clio Bilbee	3.29	3
Queen and I, The (Drottningen och jag)	Viviana Corrett	8.99	6244
Highly Dangerous	Riobard Solloway	4.49	42
Green Tea (Lü cha)	Julia Giovannetti	14.99	249
After Alice (Eye of the Killer)	Lil Itzkovwich	34.99	9806
Meteor	Billie Lovatt	5.49	42

Task 2: Create a URL '/books/lab8/task2', HTML file, and view function to list books that have editions higher than three and either the title or author of the book contains the two adjacent letters 'qu' by combining two or more Q operator (through "&" or " | ").

Books with edition > 3 and title or author contains "qu"

Title	Author	Price	Edition
Midnight Movie	Miquela Struys	22.99	7
Queen and I, The (Drottningen och jag)	Viviana Corrett	8.99	6244
Quinceañera	Nonna Klausewitz	6.99	2121
When the Levees Broke: A Requiem in Four Acts	Kirby Kamienski	59.99	28908

Task 3: (opposite to Task 2) Create a URL '/books/lab8/task3', HTML file, and view function to list books that have no editions higher than three and either the title or author of the book that does not contain the two adjacent letters "qu" by combining two or more Q operator (through "&" or "|", beside ~ operator).

Books with edition ≤ 3 and no "qu" in title or author

Title	Author	Price	Edition
Continuous Delivery	J.Humble and D. Farley	120.0	3
Continuous Delivery	J.Humble and D. Farley	0.0	1
Networks and Security	Ali Ahmad	150.0	3
Applause (Applaus)	Von Lagen	1.99	1
Perfect Blue	Marieann Liston	69.99	2
We Are Young. We Are Strong.	Clio Bilbee	3.29	3

Task 4: Create a URL '/books/lab8/task4', HTML file, and view function to list books and order by their titles using the function "order_by".

A Blank on the Map	Nevile Suthworth	3.79	5333
A Cinderella Story: Once Upon a Song	Codee Cater	2.89	8417
After Alice (Eye of the Killer)	Lil Itzkovwich	34.99	9806
Applause (Applaus)	Von Lagen	1.99	1
Beauty in Trouble (Kráska v nesnázích)	Rockey Doucette	8.99	74
Carousel	Mordy Skingle	24.99	285
Castle Freak	Worthington Fairleigh	19.99	7
Continuous Delivery	J.Humble and D. Farley	120.0	3
Continuous Delivery	J.Humble and D. Farley	0.0	1
Curious Case of Benjamin Button, The	Estevan Kleinzweig	19.99	739
Faculty, The	Dorey Fraulo	5.49	40457
Getaway	Maxy Wilton	24.99	582
Good Son, The	Elnore Woodroff	19.99	69
Green Tea (Lü cha)	Julia Giovannetti	14.99	249
Highly Dangerous	Riobard Solloway	4.49	42
Hothead (Coup de tête)	Keslie Ivashkin	2.49	389
If I Want to Whistle, I Whistle (Eu cand vreau sa fluier, fluier)	Nicky Lukash	18.99	166
Jackass 2.5	Veradis Jarmain	3.69	84422
Lawyer Man	Rancell Wissbey	39.99	694
Living Room of the Nation, The (Kansakunnan olohuone)	Gwenore Selvey	39.99	781
Love, Wedding, Marriage	Heath Lauxmann	25.0	166
Meteor	Billie Lovatt	5.49	42
Midnight Movie	Miquela Struys	22.99	7
Networks and Security	Ali Ahmad	150.0	3
Oh, God! Book II	Elayne Mantle	3.79	99
Patent Leather Kid, The	Reinald Jeffes	19.99	285
Perfect Blue	Marieann Liston	69.99	2

Task 5: Create a URL '/books/lab8/task5', HTML file, and view function to display the number of books, total price of all books, average price, maximum price, and minimum price using aggregation functions.

- Total number of books: 45
- Total price: 1463.19
- Average price: 32.51533333333334
- Maximum price: 349.99
- Minimum price: 0.0

Task 6: Create a database scheme (through models in Django)

Note that instead of creating another app to handle student applications, you may exploit existing apps, like user or book modules

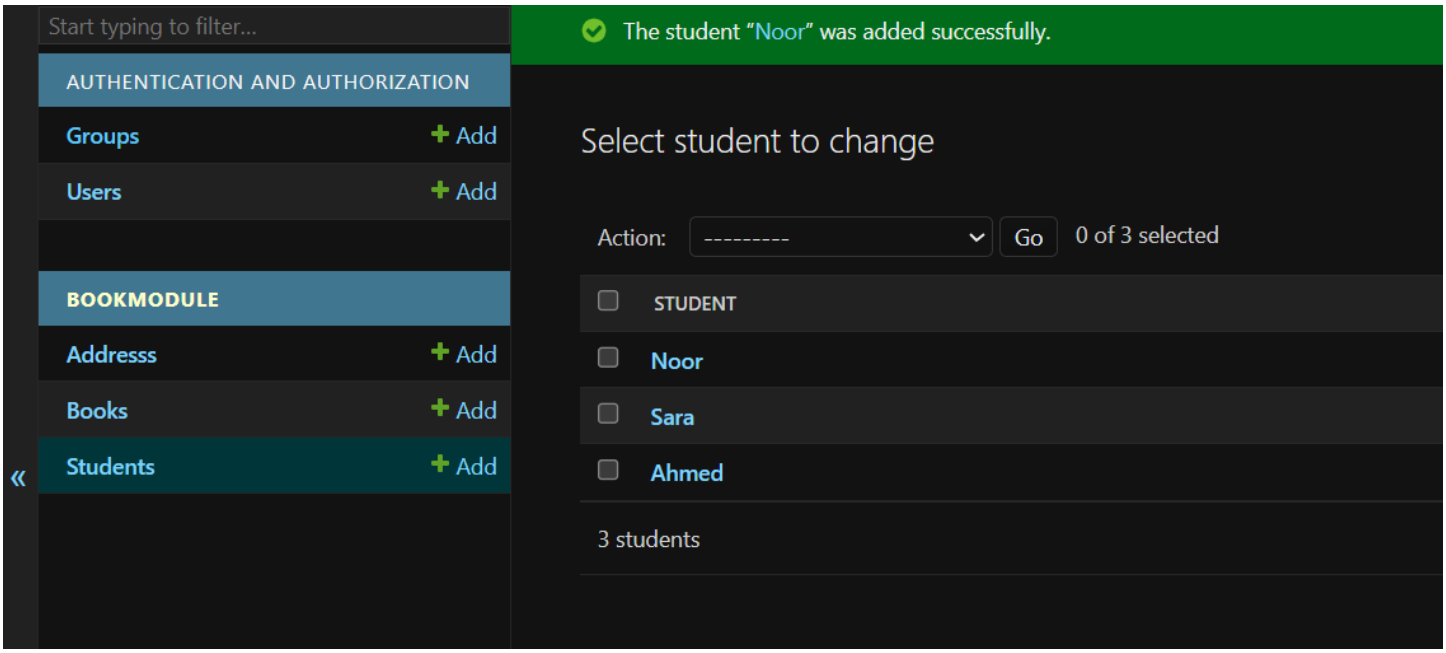
Student

ID	as an integer (this is done automatically by Django)
name	as a string
age	as an integer

address	as a foreign key to table address

Address

ID	as an integer (this is done automatically by Django)
city	as a string



Task7: Create a page that shows the number of students in each city.

Number of students in each city

City	Number of Students
Riyadh	2
Jeddah	1
Buraydah	0